

VOLCANISM AND TECTONICS ON VENUS

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KEY WORDS: mantle convection, thermal history, elastic thickness, lithosphere, gravity

ABSTRACT

We review recent developments in the study of volcanism and tectonics on Venus. Venus's crust is basaltic, dry, and probably about 30 km thick. The mantle convects, giving rise to plumes, and has a similar composition and mean temperature ($\sim 1300^\circ\text{C}$), but a higher viscosity ($\sim 10^{20}$ Pa s), than that of the Earth. Inferred melt generation rates constrain the lithospheric thickness to between 80 and 200 km. The elastic thickness of the lithosphere is about 30 km on average. The present-day lack of plate tectonics may be due to strong faults and the high viscosity of the mantle. Most of the differences between Earth and Venus processes can be explained by the absence of water.

Venus underwent a global resurfacing event 300–600 Ma ago, the cause and nature of which remains uncertain. The present-day surface heat flux on Venus is about half the likely radiogenic heat generation rate, which suggests that Venus has been heating up since the resurfacing event.

INTRODUCTION

Over five years ago, the NASA spacecraft Magellan began sending back information from Venus (Saunders et al 1992). During this time, a great deal of work concerning crustal and tectonic processes on the planet has been produced, much of which has been reviewed elsewhere (Phillips & Hansen 1994, Kaula 1994, Bindschadler 1995). The aim of this article is to synthesize much of the recent work and to show that a consistent description of present-day geological processes on Venus can be produced. Parts of this description are not in dispute; others, however, are still controversial.

Summary of Present-Day Processes on Venus

The previous sections show that some processes on Venus are similar to those on Earth and others are different. Considering the similarities first, Venus has a basaltic crust, an average effective elastic thickness of about 25–30 km, a mechanical boundary layer thickness probably around ~ 100 km, and a mantle potential temperature of about 1300°C , all of which are similar to terrestrial oceanic lithospheric characteristics. As with Earth, mantle convection is vigorous ($Ra \sim 10^7$) and produces plumes (which are probably time dependent) that cause uplift, rifting, and volcanism.

Considering the differences, the surface appearance of Venus is dissimilar to Earth's because the lack of water has greatly reduced erosion and sedimentation and also because of the apparent absence of plate tectonics (at least since the resurfacing event 300–600 Ma ago). The viscosity of the mantle beneath the lithosphere on Venus is probably an order of magnitude higher than that on Earth, and the surface heat flux on Venus is about half the heat generation rate. Venus is therefore currently heating up, whereas Earth is cooling down, which implies that the thermal histories of the two planets must be different. The crustal thickness of Venus is uncertain, but it is probably about 30 km. The morphology of rifts such as Devana Chasma suggests that faults on Venus are capable of supporting stresses about eight times greater than faults on Earth.

The atmosphere of Venus is extremely dry, perhaps because of a different impact history during planetary formation. Dry rocks retain greater strength at high temperatures than do wet rocks, and dry melts have viscosities that are orders of magnitude higher than those of wet melts. Dry melting occurs at higher temperatures than wet melting, and phase transitions happen much more slowly in the absence of water. Because pore fluid pressure reduces the shear stresses that faults can withstand, faults under dry conditions are stronger than faults in wet conditions.

Assuming that the mantle of Venus is as dry as the atmosphere, and has been throughout the last 0.5 Ga, then the following observations can all be explained simply by the absence of water: high mantle viscosity beneath the lithosphere; fault strength that is about eight times that on Earth; and similar T_e to terrestrial oceanic lithosphere (despite a surface temperature of 450°C).

Perhaps surprisingly, most of the differences between Earth and Venus can be explained by one simple factor. The main conclusion of this review is that the fundamental importance of water to processes on Earth has not been fully recognized.

EVOLUTION OF GEOLOGICAL PROCESSES

The previous section shows that a consistent description of present-day processes on Venus can be developed. As with Earth, however, the route by which

Venus reached its present-day state is much less well understood. This section discusses four related topics: first, what little has been deduced about the history of Venus from the surface record; second, why Venus does not currently possess plate tectonics; third, the nature of the resurfacing event; and finally, how all these topics may relate to the thermal history of the planet.

Surface Record of Evolution

A knowledge of a planet's stratigraphy is a prerequisite for understanding its evolution, which in turn requires good age data. On Venus, relative age relationships may be obtained by normal geological observations, but absolute age determinations depend on cratering counts.

Regional examinations of overlapping relationships have as yet produced little information, although a global stratigraphic survey (Basilevsky & Head 1995) concluded that the resurfacing event involved tectonic and volcanic activity, with declining surface volcanism thereafter. Pre-resurfacing units may survive in tessera regions. At least in some areas, the emplacement of volcanic plains was geologically instantaneous over areas of up to 10^7 km² (Basilevsky & Head 1996).

Despite the possible variations in surface age discussed above, >80% of the surface is likely to have had all previous craters removed in a relatively short time compared to 500 Ma, probably between 10 and 100 Ma (Phillips et al 1992, Strom et al 1994, Namiki & Solomon 1994). Only the last episode of resurfacing is recorded: Whether it was a single event or a process that finally stopped over a 10–100 Ma period is impossible to tell. Following the last resurfacing, based on the small number of craters that are volcanically embayed from outside or tectonized, less than 20% of the planet was resurfaced over the next 300–600 Ma (Schaber et al 1992, Namiki & Solomon 1994, Strom et al 1994). The nature of the resurfacing event is discussed below.

Plate Tectonics

Previous authors have recognized that plate tectonics on Earth can take place because the driving forces (especially slab pull) exceed the resistive forces (primarily fault friction, flexure, and viscous drag) (e.g. Forsyth & Uyeda 1975, McKenzie 1977). The apparent lack of plate tectonics on Venus is of interest, and the balance of forces is likely to be different.

One possible explanation for the present-day lack of plate tectonics on Venus is that the high surface temperature and thick basaltic crust make the lithosphere less dense than the underlying convecting mantle, thereby inhibiting subduction. The higher surface temperature means that the average lithospheric temperature is about 200°C higher than that on Earth, assuming its base is defined by the same isotherm. This temperature difference alone would reduce the density

Subduction = Abtauchen der Lithosphäre unter tek. platte

may exert sufficient force locally to cause plate motion, but only on a plume length scale (~ 1000 km) rather than on the 10,000-km scale seen on Earth.

The Nature of the Resurfacing Event

Virtually the only constraints on the resurfacing event come from the crater record (see above), which suggests that the last global resurfacing took place in less than 100 Ma and was followed by reduced levels of activity. In addition, if the conclusions regarding the present heat flux are correct, then Venus must have been heating up since the resurfacing event because the present rate of heat loss is less than the inferred radiogenic heat generation rate.

Various models have been suggested to explain the above observations, many of which require catastrophic events (see below). Rather than investigate these suggestions further, it is instructive to examine two potential resurfacing mechanisms that are constrained by terrestrial observations, namely resurfacing by plate tectonics and resurfacing by plume activity.

PLATE TECTONIC RESURFACING Plate tectonics, which is taken to mean the creation of lithosphere at spreading ridges and its destruction at trenches, could certainly account for the rapid resurfacing. Assuming that the last resurfacing event took 100 Ma to complete, the average surface age during this time must have been 50 Ma. This age is similar to the average age of oceanic lithosphere on Earth (Sclater et al 1980).

The main topographic features of plate tectonics—ridges, trenches, and transform faults—tend to disappear once plate tectonics has stopped, either because they are no longer thermally maintained or because they are topographic lows that are likely to be preferentially filled in by volcanism. Other characteristics of plate tectonics, such as linear bands of faulting, are also likely to be obscured over the succeeding 500 Ma. Thus, the lack of evidence for present-day plate tectonics on Venus cannot be used as evidence that it was not the cause of the resurfacing event. However, it is not clear whether plate motions could stop within the 100-Ma time scale required by the cratering record.

PLUME-RELATED RESURFACING The other suggested mechanism of resurfacing is plume melt generation. Because less than 20% of the planet has been resurfaced over the last 500 Ma, whereas the resurfacing event took less than 100 Ma, the global surface melt production rate must have been a factor of at least 25 higher than the present-day rate for the 100 Ma before the resurfacing event ended. Such a rate of melt generation could be achieved by increasing the number of plumes, increasing the potential temperature by about 200°C (see Figure 3c), reducing the lid thickness, or reducing the solidus temperature (probably by the addition of water). However, it is not clear that any of these mechanisms can produce a rapid reduction in melt generation over 100 Ma: For instance, it is difficult to see how the plume population could decrease by 96%